

Ayush Kumar Singh

Phone: +91-8797781945

Email: ayushks1602@gmail.com

GitHub: github.com/AyushK5ingh
LinkedIn: linkedin.com/in/ayushk5ingh/
Website: ayushk5ingh.tech/

EDUCATION

Ajay Kumar Garg Engineering College

B.Tech in Information Technology

Ghaziabad, Uttar Pradesh

Nov 2022 - May 2026

EXPERIENCE

Nestle India

Full-Stack AI ML Engineer Intern

Gurugram, Haryana

Apr 2025 - Jul 2025

- Built **responsive full-stack web applications** using **React.js**, **Node.js**, **Express.js**, and **RESTful APIs**, creating interactive dashboards for real-time data monitoring visualization, reducing manual reporting time by **60%**.
- Engineered **backend services and API endpoints** with data validation logic, processing large datasets with optimized database queries and achieving **99%+ system accuracy**.
- Implemented **automated reporting modules** using Python and openpyxl, and integrated **unit/integration testing** with Jest across the application stack.

TECHNICAL SKILLS

Frontend: React.js 18/19, Next.js 14/15, TypeScript, JavaScript ES6+, Redux, HTML5, CSS3, Tailwind CSS, Responsive Design

Backend: Node.js, Express.js, Python, GO, GIN, Golang, RESTful Application Programming Interfaces (APIs), GraphQL, tRPC, WebSockets (Socket.io), Prisma Object-Relational Mapping (ORM), Authentication (JWT, OAuth)

Databases: PostgreSQL, MySQL, MongoDB, Redis, Database Design, Query Optimization

DevOps & Tools: AWS (EC2, S3, Lambda), Docker, CI/CD GitHub Actions, Git, VS Code, Postman, npm/yarn

Core Concepts: Data Structures & Algorithms, System Design, Microservices, Object-Oriented Programming (OOP), Agile/Scrum, Software Development Life Cycle (SDLC), Testing (Jest, Test-Driven Development)

PROJECTS

Vero — Full-Stack Development Platform

[Live Demo](#) — [GitHub](#)

Tech Stack: Next.js 15, React 19, TypeScript, Node.js, PostgreSQL, Docker, tRPC, Prisma

- Built a **production-grade full-stack application** with **Next.js 15**, **React 19**, and **TypeScript**, implementing Server-Side Rendering (SSR), Static Site Generation (SSG), and API routes for optimal performance and Search Engine Optimization (SEO).
- Designed a **type-safe backend API layer** using **tRPC** and **Prisma ORM**, enabling end-to-end type safety and reducing API-related bugs by **40%**.
- Engineered **microservices architecture** with containerized services using Docker, implementing CI/CD pipelines with GitHub Actions and integrating third-party APIs with rate limiting and caching.

Synthex — Real-Time Collaborative Code Editor

[Live Demo](#) — [GitHub](#)

Tech Stack: Next.js 15, TypeScript, Node.js, Socket.io, Redis, MongoDB, Judge0 API, AWS S3

- Created a **real-time collaborative editor** using **Next.js**, **TypeScript**, and **Socket.io**, featuring multi-user cursor tracking, syntax highlighting, and code execution across **15+ languages**.
- Architected a **scalable WebSocket backend** using **Socket.io** and **Redis Pub/Sub**, enabling **500+ concurrent sessions** with efficient state synchronization and conflict resolution.
- Implemented **RESTful APIs** with Express.js/MongoDB for authentication and project management, integrated Razorpay payment processing, and achieved **20% faster page loads** through optimization techniques.

OPEN SOURCE CONTRIBUTIONS & ACHIEVEMENTS

- **Meshery Open Source (CNCF):** Contributing to cloud-native service mesh management platform with merged PRs for feature enhancements and bug fixes using Kubernetes and Go.
- **Certifications:** Software Engineering Job Simulation (JPMorgan Chase), Full Stack Web Development (Sheryians), Data Science with Python and OOPs with Python (Infosys Springboard).
- Solved 250+ algorithmic problems on LeetCode and CodeChef using C++